

SIMATS ENGINEERING

CO4 – AT1 - Article Review- Low

COURSE NAME: Advance Wireless Communication Systems /
5G / 6G Communication Systems
COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 5 - *Develop 5G-enabled edge computing and IoT solutions for smart city, healthcare, industrial automation, and intelligent communication applications. (BL6)*

Assessment Tool Weightage: 25%

Short description about assessment: Students review recent technical articles to assess critical thinking and awareness of trends.

Dr.

QUESTIONS: Total Marks: Duration: 90 Minutes

1. Multi-access Edge Computing (MEC) Architectures for Low-Latency 5G Applications (BL5)
2. Role of MEC in Ultra-Reliable Low-Latency Communications (URLLC) (BL5)
3. Cloud Radio Access Network (Cloud RAN): Architecture, Benefits, and Challenges (BL5)
4. Virtualization and Resource Management in Cloud RAN (BL5)
5. Edge AI for Intelligent 5G Network Optimization (BL5)
6. Artificial Intelligence-Based Resource Allocation in Edge Computing (BL5)
7. Integration of MEC and Artificial Intelligence for Next-Generation Networks (BL5)
8. Security and Privacy Challenges in Edge Computing for 5G (BL5)
9. Dynamic Spectrum Sharing Techniques for Efficient Spectrum Utilization (BL5)
10. AI-Driven Dynamic Spectrum Sharing in 5G and Beyond Networks (BL5)
11. Satellite Backhaul Solutions for Rural and Remote 5G Connectivity (BL5)
12. Low Earth Orbit (LEO) Satellite Networks for 5G and 6G Communications (BL5)
13. Comparative Analysis of LEO, MEO, and GEO Satellites for Wireless Communication (BL4)
14. Integration of Terrestrial and Non-Terrestrial Networks (NTN) in 6G (BL5)
15. Edge Computing over Satellite Networks for Future Wireless Systems (BL5)

16. NB-IoT Architecture and Emerging Industrial Applications (BL5)
17. LTE-M Technology for Massive Machine-Type Communications (BL5)
18. Comparative Study of NB-IoT and LTE-M for IoT Deployments (BL4)
19. Power-Efficient Communication Techniques for IoT Devices in 5G Networks (BL5)
20. Battery Life Optimization Strategies for Massive IoT Devices (BL5)
21. Wearable Healthcare Devices Enabled by 5G Communication (BL5)
22. Smart Sensors for Intelligent IoT Applications in 5G Networks (BL5)
23. Energy-Efficient Communication Protocols for Large-Scale IoT Systems (BL5)
24. Intelligent Traffic Management Using 5G and Edge Computing (BL5)
25. AI-Enabled Smart Surveillance Systems over 5G Networks (BL5)
26. Smart Energy Monitoring and Management Using IoT and 5G (BL5)
27. Digital Twin Technology for Smart City Infrastructure (BL5)
28. Edge AI Applications for Sustainable Smart Cities (BL5)
29. Industrial IoT Applications in Smart Manufacturing (BL5)
30. Predictive Maintenance Using Artificial Intelligence and 5G Networks (BL5)
31. Robotic Automation in Industry 4.0 Using 5G Communication (BL5)
32. Private 5G Networks for Smart Factory Deployments (BL5)
33. Edge Computing for Real-Time Industrial Automation (BL5)
34. Remote Robotic Surgery Enabled by 5G Communication Networks (BL5)
35. AI-Driven Medical Diagnosis Using 5G and Edge Computing (BL5)
36. Intelligent Remote Patient Monitoring Using Wearable IoT Devices (BL5)
37. Secure Healthcare IoT Systems in 5G Networks: Challenges and Solutions (BL5)
38. Smart Irrigation Systems Using IoT and 5G Technologies (BL5)
39. Precision Agriculture for Crop Monitoring and Yield Prediction Using AI and 5G (BL5)
40. Intelligent Livestock Tracking and Smart Farming Using 5G IoT Networks (BL5)

Code of Conduct:

I B.NANI GANGA PRASAD(192412398) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Article Review					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

DESCRIPTION:

Cloud Radio Access Network (Cloud RAN or C-RAN) is an advanced architecture for modern wireless communication networks that focuses on centralization, virtualization, flexibility, and efficient resource utilization. The reference journal discusses the evolution of RAN architectures toward centralized and cloud-native approaches, particularly in the context of O-RAN. In C-RAN, radio-related functions are separated from centralized processing resources, allowing multiple network components to share common computing infrastructure. This architecture provides greater flexibility in managing network resources and creates a suitable foundation for advanced 5G network services.

One of the major benefits of C-RAN is centralized resource management and scalability. Centralized processing allows network data from different sources and network functions to be collected, standardized, stored, and processed using a common platform. The reference journal describes how cloud-native technologies support data acquisition, processing, storage, visualization, automation, and AI/ML applications across multi-vendor RAN environments. Such an architecture can improve resource utilization, simplify network management, support automation, and make it easier to scale network resources as requirements increase.

Despite its advantages, C-RAN faces challenges related to high-performance connectivity, data management, latency, synchronization, security, and scalability. Large volumes of network data must be collected and transformed before they can be effectively used for analytics and intelligent applications. The journal highlights that raw RAN data often requires validation, enrichment, transformation, and standardization before it can support AI-based applications. Therefore, efficient cloud infrastructure, reliable communication between radio and centralized processing functions, and automated data-management techniques are important for successful C-RAN deployment. Overall, C-RAN provides an important foundation for developing flexible, saleable, efficient, and intelligent 5G and future wireless networks.



Contents lists available at ScienceDirect

Engineering Applications of Artificial Intelligence

journal homepage: www.elsevier.com/locate/engappai

Anomaly detection in offshore open radio access network using long short-term memory models on a novel artificial intelligence-driven cloud-native data platform[☆]

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ARTICLE INFO

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ABSTRACT

The Radio Access Network (RAN) is a critical component of modern telecommunications infrastructure, currently evolving towards disaggregated and open architectures. These advancements are pivotal for integrating intelligent, data-driven applications aimed at enhancing network reliability and operational autonomy through the introduction of cognitive capabilities, as exemplified by the emerging Open Radio Access Network (O-RAN) standards. Despite its potential, the nascent nature of O-RAN technology presents challenges, primarily due to the absence of mature operational standards. This complicates the management of data and intelligent applications, particularly when integrating with traditional network management and operational support systems. Divergent vendor-specific design approaches further hinder migration and limit solution reusability. These challenges are compounded by a skills gap in telecommunications business-oriented engineering, which remains a key barrier to effective O-RAN deployment and intelligent application development. To address these challenges, Boldyn Networks developed a novel cloud-native data analytics platform, specifically designed to support scalable Artificial Intelligence (AI) integration within O-RAN deployments. This platform underwent rigorous testing in real-world scenarios, and applied advanced AI techniques to improve operational efficiency and customer experience. Implementation involved adopting Development Operations (DevOps) practices, leveraging data lakehouse architectures tailored for AI applications, and employing sophisticated data engineering strategies. The platform successfully addresses connectivity challenges inherent in real-world offshore wind farm deployments using Long Short-Term Memory (LSTM) models for anomaly detection in network connectivity. After integrating the LSTM models into the network control, more than 90 percent of connectivity issues were reduced in runtime. This marks a step toward autonomous, self-organizing, and self-healing networks.

1. Introduction

Telecommunication networks are essential to many aspects of our lives, driving digital transformation and revolutionizing communication. The benefits of these networks are numerous. Recently, their importance has surged due to the proliferation of various types of user equipment (UE), internet of things (IoT) devices, autonomous operations, and services that require faster, more reliable, resilient, secure, and private connectivity. This has led to a substantial rise in the demand for private networks, amplifying the challenges of managing many smaller, tailored mobile networks to deliver high-quality services.

To address these escalating demands, innovative enhancements in network design have emerged. A recent progress in this arena is the advent of open radio access network (O-RAN) technology. O-RAN aims to disaggregate the monolithic, single-vendor RAN reducing infrastructure costs and vendor lock-in, and paving the way for network programmability, ultimately leading to autonomous network operations by leveraging data and native-supported in-network artificial intelligence (AI) techniques, aiming to streamline the complexities of designing, delivering, managing and operating private networks. It

[☆] This work is a continuation of the research conducted within the Innovate UK/CELTIC-NEXT European collaborative project AIMM (AI-enabled Massive MIMO).

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